



Bread contamination detection using deep learning and thermal imaging

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ABSTRACT

The effectiveness of four You Only Look Once (YOLO) models (YOLOv5, YOLOv8, YOLOv9, and YOLOv11) was evaluated for detecting contamination (water, oil, grease, and cleaning chemicals) in bread using thermal imaging. Thermal images of contaminated and uncontaminated bread slices were collected through active thermography, employing a controlled hot air supply under varying conditions. The analysis revealed distinct thermal signs, with oil-contaminated areas exhibiting higher temperatures, and water, grease, and chemical contaminants appearing cooler. The trained YOLO models were assessed based on the mean average precision (mAP50-95), inference speed, and computational efficiency. The results indicate that increased model complexity does not always translate into higher accuracy. The lightweight YOLOv11n model (2.59 M parameters, 6.4 Giga Floating Point Operations Per Second (GFLOPs)) achieved a competitive mAP50-95 score of 0.607, closely matching larger models such as YOLOv8s (28.6 GFLOPs, mAP50-95: 0.601) and YOLOv9s (27.4 GFLOPs, mAP50-95: 0.601). Despite deeper architectures, models such as YOLOv9s exhibit signs of overfitting without significant performance improvements. YOLOv11n outperformed the other models in terms of operational efficiency, achieving faster inference speeds (3.0 ms/image) and lower memory consumption, thus making it suitable for real-time industrial applications. The training durations for all models were low, ranging between 0.107 and 0.146 h for 10 epochs on an NVIDIA Tesla T4 GPU, demonstrating that contamination detection is a computationally manageable task. These findings suggest that lightweight models, such as YOLOv11n, provide an optimal balance between accuracy, computational efficiency, and practical deployment feasibility.

1. Introduction

Globally, bread is a widely consumed basic food item on a daily basis (Cacak-Pietrzak et al., 2023). This widespread popularity can be attributed to its nutritional value and sensory qualities (Mollakhalili-Meybodi et al., 2023). The global bread industry is estimated to account for approximately US \$ 335 billion by 2023. The compound annual growth rate of 2.6 % from 2024 to 2030 to, is driven by the increasing demand for convenient and healthier options, including whole-grain and gluten-free products (“Packaged Bread Market Size, Share & Growth Report, 2025,”). There are several stages of bread processing, from mixing and kneading to baking and packaging, during which contamination can occur (Edwige, n.d.; Turan et al., 2024). Contaminants such as water, oil, grease, and cleaning chemicals endanger the quality and safety of bread and have proven to be a major

menace to manufacturers (Maher and Nowak, 2022). In the baking industry, food hygiene and safety should be ensured by ensuring that food products are free from foreign substances, as this can lead to regulatory challenges and reduce customer trust in the products (Okpala and Korzeniowska, 2023). Unintended physical (metal fragments, plastic pieces, glass shards, stones or sand, insects), biological (mold, Salmonella, E. coli, Listeria, yeast overgrowth, fungal contamination), and chemical (cleaning chemicals, grease and lubricants, oil, contaminated water, pesticide residues) contaminants entering food production are classified as foreign materials requiring mitigation (Mutlu et al., 2025; Payne et al., 2023).

Studies have indicated that approximately 15 % of bakery-related food safety incidents are caused by cross-contamination during cleaning operations (Gargiulo et al., 2022; Soon et al., 2020). If these foreign substances remain unchecked, they can pose safety risks and question

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the integrity of sliced bread, thereby contradicting the regulatory standards (Cholis et al., 2025). Contamination detection in bread encompasses a spectrum of methodologies ranging from basic sensory evaluations to advanced machine-learning approaches (Görgülü, 2025). Fundamental techniques include visual inspection (Treepong and Theera-Ampornpant, 2023), microbial testing (Sasikala et al., 2023), pH measurements (Su et al., 2019), and moisture content analysis (Madasamy Raja et al., 2025). Intermediate methodologies include UV light analysis (Romano et al., 2024) and spectroscopic techniques for detecting chemical and microbial contaminants (Hernandez-Aguilar et al., 2020). Advanced techniques, such as hyperspectral imaging (Soni et al., 2022), X-ray scanning (Sala et al., 2024), and deep learning models CNN, YOLO, contribute to the precise identification of surface and physical contaminants (Chen et al., 2025; Kılıcı and Koklu, 2025; Sun et al., 2022). The integration of biosensors and predictive analytics powered by artificial intelligence can significantly enhance food safety by detecting harmful substances and preventing contamination (Chen et al., 2024; Hassan et al., 2024; Zhang et al., 2022).

Thermal imaging has shown adequate results in terms of food safety for detecting contaminants, even under conditions of low contrast, with the aid of specialized image processing techniques (Agriculture et al., 2022; Pathmanaban et al., 2023b). Thermal imaging detects foreign bodies in food by exploiting differences in their thermal properties. Compared to other object detection models, YOLO-based detectors offer a unique combination of speed and accuracy owing to their single-stage architecture, which is ideal for real-time applications such as industrial food inspection (Dhelia et al., 2024). Prior research in the food sector has demonstrated YOLO's superior performance of YOLO in identifying surface-level defects, achieving high detection accuracy even under challenging conditions, such as low contrast or mixed contamination types. (Hussain, 2023). Analyzing cooling behavior, especially with active thermography, creates high-contrast images, enabling contaminant identification and improved food safety (Pugazhendi et al., 2023). In addition, thermal imaging combined with CNN has been used to classify and quantify adulterated minced mutton. This approach achieved high accuracy (99.97 %–99.99 %) in classification and strong predictive performance ($R^2 = 0.9933$) for adulteration levels (Zheng et al., 2021). Furthermore, these techniques have been applied to microbial contamination detection in meat, fungal growth monitoring in fruits (Barmettler et al., 2025), and seal contamination detection in food packaging (Singh et al., 2025), among other applications. In addition to their detection capabilities, YOLO models generate precise bounding boxes that allow for the quantification of contamination areas (Jayakrishna and Ganesh, 2025). This is essential for assessing the severity of contamination or classifying samples according to contamination levels. Furthermore, YOLO models can be trained to distinguish different types of contaminants, such as water, oil, grease, and cleaning agents, by learning their unique thermal profiles, thereby enhancing the model's applicability in food safety monitoring (Djedidi et al., 2025). Thermal imaging integrated with YOLO models has shown remarkable effectiveness in food industry applications, YOLOv8 achieving high accuracy in broiler pathology detection (mAP50 of 0.988) and automated counting (95 % mAP) (Abdalshefie Abuhussein et al., 2024; Elmessery et al., 2023), while also demonstrating success in quality control for food spoilage detection (Dhelia et al., 2024). Additional applications include YOLOv4 implementation in aquaculture for fish deformity detection (Kuswantori et al., 2022) and MSD-YOLO development for industrial surface defect detection (Yan et al., 2024). To thoroughly evaluate the performance of YOLO in this task, four different YOLO variants, YOLOv5s, YOLOv8s, YOLOv9s, and lightweight YOLOv11n, were selected. This comparative approach helps analyze the trade-offs between model complexity, detection accuracy, and suitability for deployment in real-time or edge computing environments, particularly in food industry settings.

To the best of our knowledge, no previous study has employed integrated thermal imaging and YOLO models to detect bread

contamination. This study aimed to fill this gap by (1) developing an extended thermal image dataset of bread samples contaminated with water, oil, grease, and cleaning chemicals; (2) performing a thorough thermal image analysis to evaluate temperature gradients; and (3) building and testing variants of the YOLO model to automate contamination detection (YOLOv11n, YOLOv8s, YOLOv9s, and YOLOv5s). An overview of the proposed system is shown in Fig. 1.

2. Materials and methods

2.1. Sample preparation

Commercial white bread (400 g) contains wheat flour, sugar, preservatives, and fortifying agents. The key allergens detected were wheat, milk, and soy. The bread provided a standardized nutritional content per 70 g of serving. The dimensions of the bread slices were kept constant (10 cm × 10 cm × 1 cm), and the moisture content was (35 ± 2 %). The bread was stored under controlled environmental conditions of 25 °C and 50 % RH before conducting any tests so that the texture and moisture content behaved properly. Four contaminants were employed at two levels to imitate these conditions in food-processing environments. Deionized water and refined vegetable oil were applied in volumes of 0.2 mL and 0.5 mL using calibrated micropipettes. Food-grade machinery grease was weighed at 0.1 g and 0.3 g on an analytical balance and applied with sterile spatulas. The diluted cleaning agent (5 % v/v solution) was dispensed at 0.1 mL and 0.2 mL. Triplicate runs were performed for each contamination scenario for statistical validity, and uniform distributions were achieved using each contaminant's specific sterile spreading tool. Fifty bread slices were prepared for each contaminant type, with corresponding uncontaminated controls (n = 50). After contamination, the samples underwent a 5-min stabilization period to attain equilibrium prior to analysis. The bread samples used in this study are shown in Fig. 2.

2.2. Image acquisition and setup

The image acquisition setup facilitated precise thermal imaging of contaminated and uncontaminated bread slices in a controlled environment with minimal external influences. Temperature variations over the surface of the bread were recorded using a FLIR One Pro thermal camera, capturing images with 160 × 120 pixel resolution and a thermal sensitivity of 70 mK. Illumination was minimized and thermal conditions were maintained by isolating the imaging apparatus in a custom-fabricated enclosure lined with black foam, which reduced reflections and ambient temperature fluctuations. The hot-air blower was positioned at a 45° angle to the surface of the bread to ideally apply a heat stimulus (35 °C airflow for 2 and 3 s, as indicated in previous studies) (Dong et al., 2022; Pathmanaban et al., 2023a). The bread slices were transferred onto a conveyor system with a non-reflective black surface to enhance the contrast in thermal imaging, thus facilitating differentiation between contaminated and uncontaminated areas. This configuration ensures uniform heating and reliable image acquisition to maximize the detection of thermal alterations due to contamination. Fig. 3 illustrates the image-acquisition setup.

2.3. Image annotation and dataset preparation

In this study, 650 thermal images were collected and processed to prepare a dataset for YOLO model training using Roboflow software (Dwyer, B., Nelson, J., Hansen, 2024). To improve the dataset quality and consequently improve the generalizability of the trained model, a series of preprocessing and augmentation procedures were performed. The preprocessing used for this study included auto-orientation of the images to have standardized alignment, static cropping over a 25–75 % horizontal and vertical location to focus on relevant sections, resizing to a single resolution of 640 × 640, auto-contrast adjustment based on

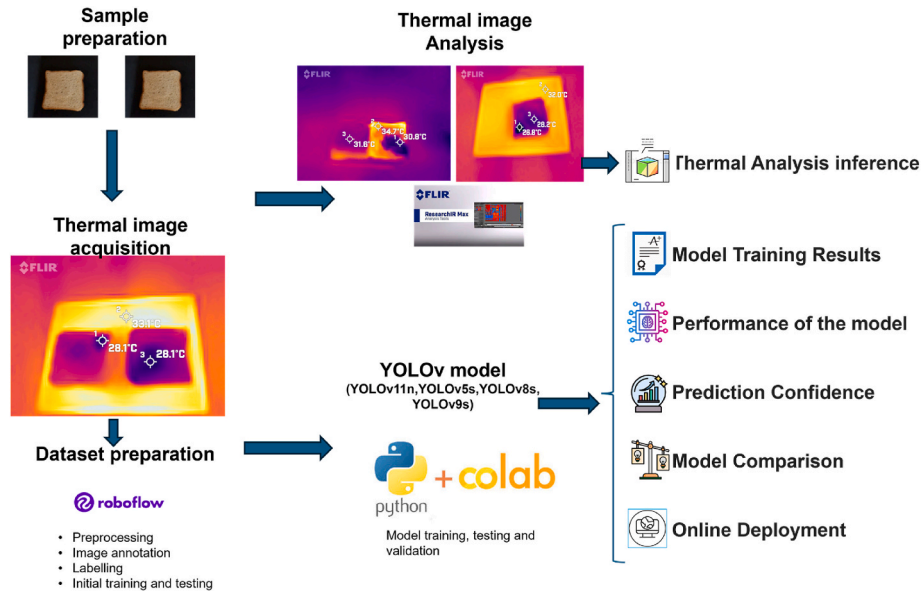


Fig. 1. Overview of proposed system.

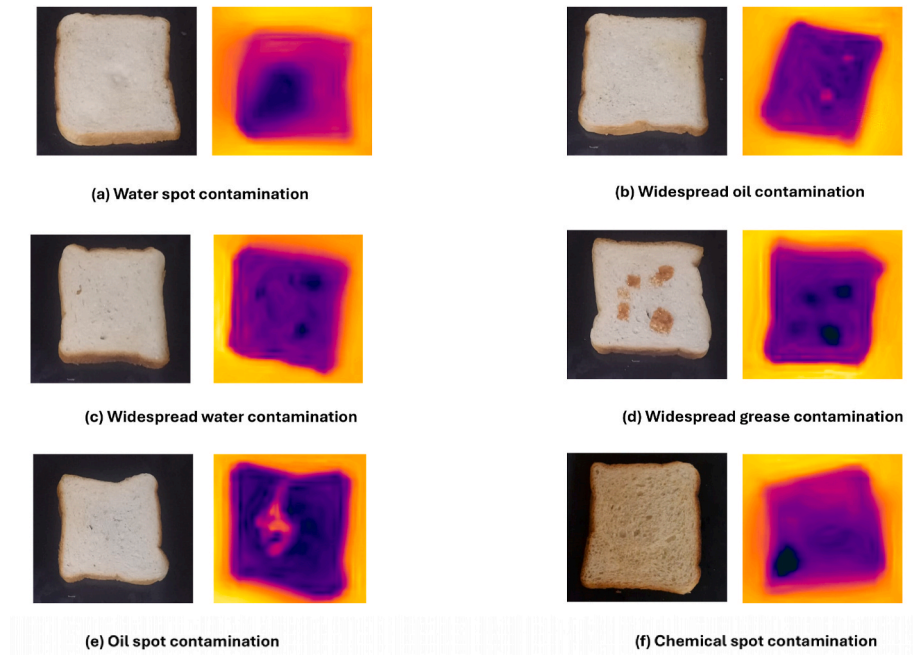


Fig. 2. Comparison of digital and thermal images of bread samples with different types of contamination. While most contaminations are not visible in digital images, thermal images reveal distinct temperature contrasts, enabling effective contamination detection.

adaptive equalization to retrieve details from the image, and conversion to grayscale to emphasize thermal variance. Additional augmentation techniques were then used to artificially widen the dataset and add variability. The augmentations included spatial diversification through horizontal and vertical flipping and changes in exposure at a scale of -10% to $+10\%$ to stimulate different lighting conditions. Thus, the original dataset was enlarged to bring the final distribution to 1342 sets of training images, 56 sets of validation images, and 28 sets of testing images, which offered a fair balance of the dataset required for vigorous training and evaluation of the models.

2.4. YOLO model selection and implementation

The YOLOv11n model was selected for real-time detection of bread contamination, balancing the need for both speed and accuracy (Khanam and Hussain, n.d.). The most recent model in the YOLO family, YOLOv11n, has incorporated state-of-the-art improvements in architecture, such as the C3k2 block for efficient feature extraction, spatial pyramid pooling fast (SPPF) for the aggregation of multiscale features, and Cross-Stage Partial with Spatial Attention (C2PSA) for enhanced attention to important areas of the image. The loss function is formulated as a combination of classification, objectness, localization, and auxiliary losses, guaranteeing correct location estimation in challenging environments including occlusions and varying scales of contamination.

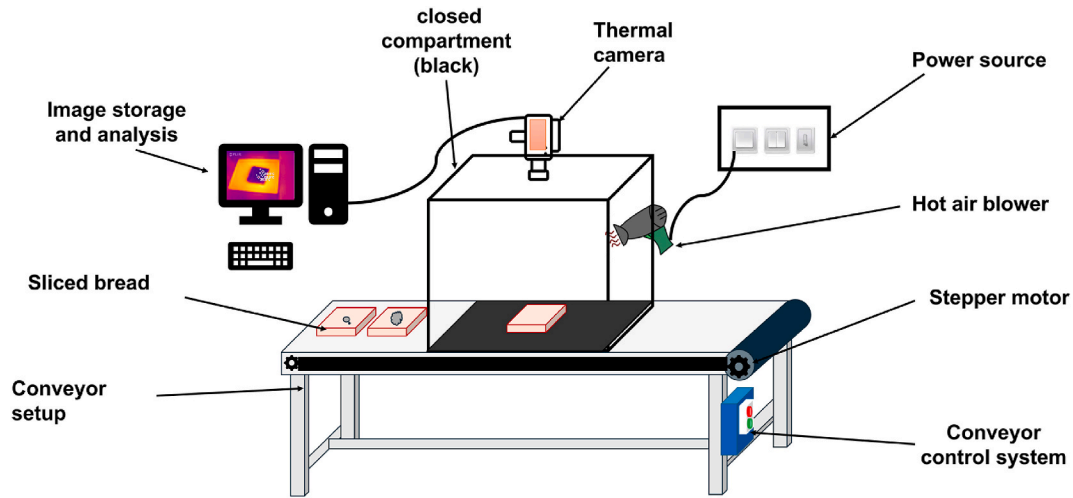


Fig. 3. Illustration of image acquisition setup.

Other models, such as YOLOv5s, YOLOv8s, and YOLOv9s, are used for comparative analysis, emphasizing the computational efficiency and accuracy advantages of YOLOv11n. The models were trained on Google Colab utilizing a Tesla T4 GPU with PyTorch 2.5.1 and CUDA 12.1, with an input resolution of 640×640 , batch size of 16, and 100 training epochs. The dataset was partitioned into several ratios (i.e., 6:2:2, 7:2:1, and 8:1:1) to observe how partitioning affects the performance of the model. The optimization was performed using stochastic gradient descent (SGD) with a learning rate of 0.01, momentum of 0.937, and weight decay of 0.0005. Mosaic transformation, horizontal flipping, and exposure adjustment augmentations added variability to the data. The model was evaluated based on a standard set of key performance index parameters, including precision, recall, mean average precision (mAP), number of parameters, and computational complexity (GFLOPs), to provide a better review of detection abilities. Fig. 4 shows the architecture of the YOLOv11n model. A detailed comparison of the different YOLO models is presented in Supplementary Materials, Table S1.

3. Result and discussion

3.1. Thermal image analysis of bread

The temperature of each contaminated bread sample was measured using thermal imaging and subsequently analyzed using the FLIR Research tool, revealing distinct thermal contrasts between the contaminated and uncontaminated regions due to variations in heat absorption and retention properties. Measurements of the surface temperature for each sample were performed in triplicate, and their means and standard deviations are reported. The temperature of uncontaminated bread sample areas appeared more uniform, with the central section averaging $28.4^\circ\text{C} \pm 0.2^\circ\text{C}$, while the surrounding area averaged $32.5^\circ\text{C} \pm 0.3^\circ\text{C}$ as shown in Fig. 5 (a). Oil spot contamination areas recorded an increase in temperature to $29.3^\circ\text{C} \pm 0.4^\circ\text{C}$ in comparison to the uncontaminated area by 1.1°C , which may be due to higher heat retention properties of oil as shown in Fig. 5 (b). This trend was repeated with oil-widespread contamination, as shown in Fig. 5(f), reflecting an increase from $29.6^\circ\text{C} \pm 0.3^\circ\text{C}$, also indicating a $+1.2^\circ\text{C}$ higher absorption property. Grease spot contamination shown in Fig. 5 (c) instead caused a substantial temperature decrease, with the contaminated area averaging $16.8^\circ\text{C} \pm 0.5^\circ\text{C}$, whereas the surroundings averaged $29.3^\circ\text{C} \pm 1.1^\circ\text{C}$, suggesting that grease acts as an insulating covering, hindering heat transfer and resulting in ΔT of -12.5°C . Water contamination also caused a significant decrease in surface temperature, with contaminated areas measuring $27.5^\circ\text{C} \pm 0.3^\circ\text{C}$, about 4.1°C lower than the surrounding uncontaminated bread, as shown in

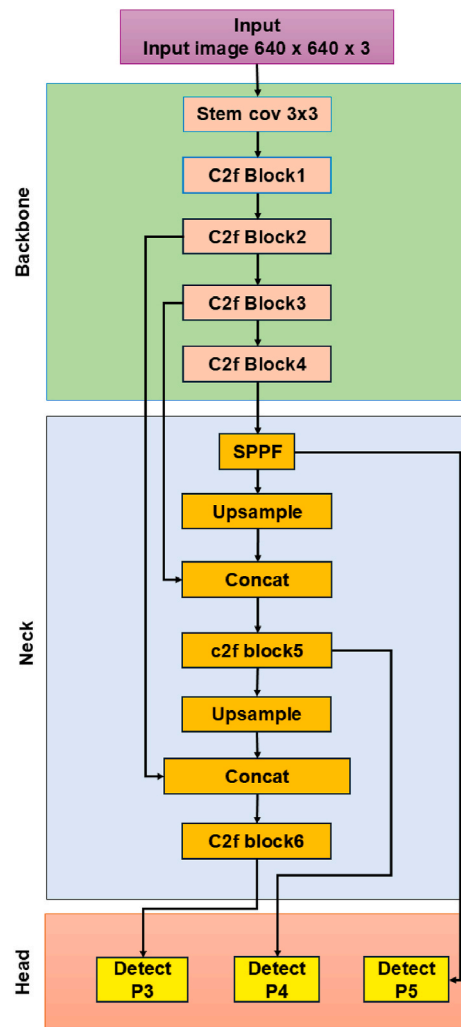


Fig. 4. YOLOv11n model architecture.

Fig. 5(e)

This can be attributed to the high specific heat capacity of water and the evaporative cooling effects. Similarly, chemical contamination produced a ΔT of -4.5°C in the contaminated regions shown in Fig. 5

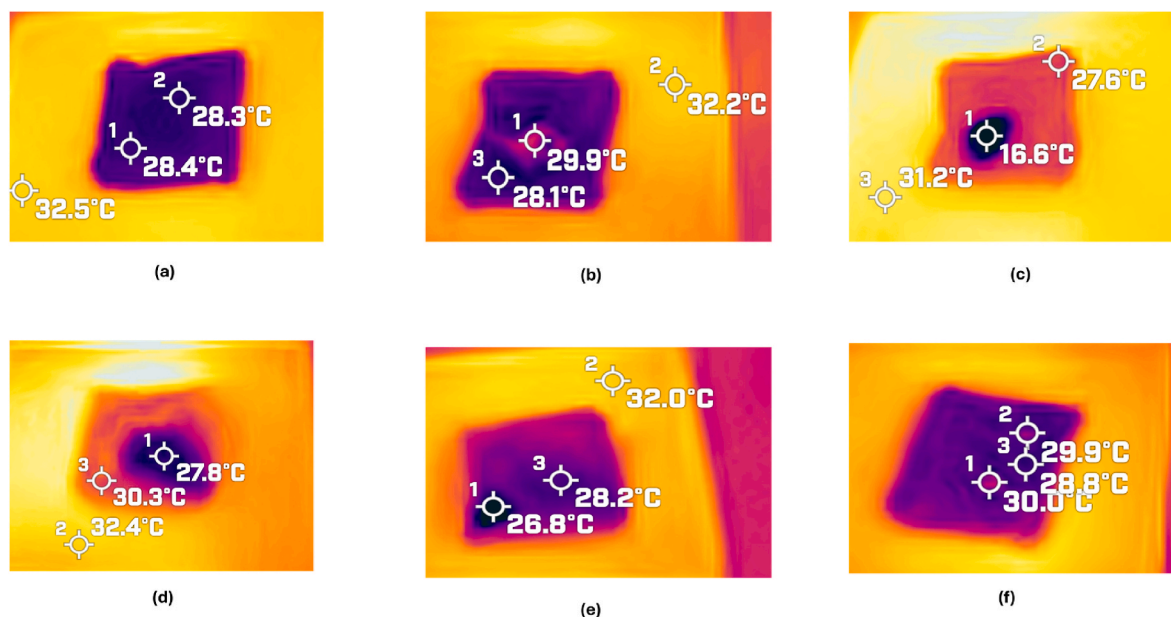


Fig. 5. Temperature analysis across contaminations (a) without contamination (b) Oil spot contamination (c) Grease spot contamination (d) Water spot contamination (e) Chemical spot contamination (f) widespread oil contamination.

(e), averaging $27.3^{\circ}\text{C} \pm 0.5^{\circ}\text{C}$, indicating that the chemical may alter the heat absorption or evaporation properties. These results indicate that contaminants, including water, grease, and chemicals, lower the surface temperature, whereas oil contaminates. Quantitative temperature analyses across contaminations are presented in [Table S2](#).

3.2. Evaluation of training results of YOLOv11n model

According to the observations in [Fig. 6](#), the training of the YOLO11n model for bread contamination detection progressed steadily for approximately 150 epochs, with the training and validation losses exhibiting a decreasing trend. Different losses, such as train/box_loss, train/cls_loss, and train/dfl_loss, reflect a constant decline, indicating a

well-learning model, while the validation losses (namely val/box_loss, val/cls_loss, and val/dfl_loss) show some fluctuation, which may imply slight overfitting, but in general stable convergence. This would be proved by performance metrics, with precision and recall converging at approximately 0.9, indicative of high classification performance. The mean Average Precision at 50 % IoU (metrics/mAP50(B)) remains high, thus showing good detection performance, while the metrics/mAP50-95 (B) steadily rises to stabilize just below 0.6, indicating an improvement in localization capabilities. Therefore, the results suggest that the YOLO11n model was successful in reliably detecting bread contamination, although some minor fluctuations in validation losses could indicate that further fine-tuning or regularization approaches might improve generalization.

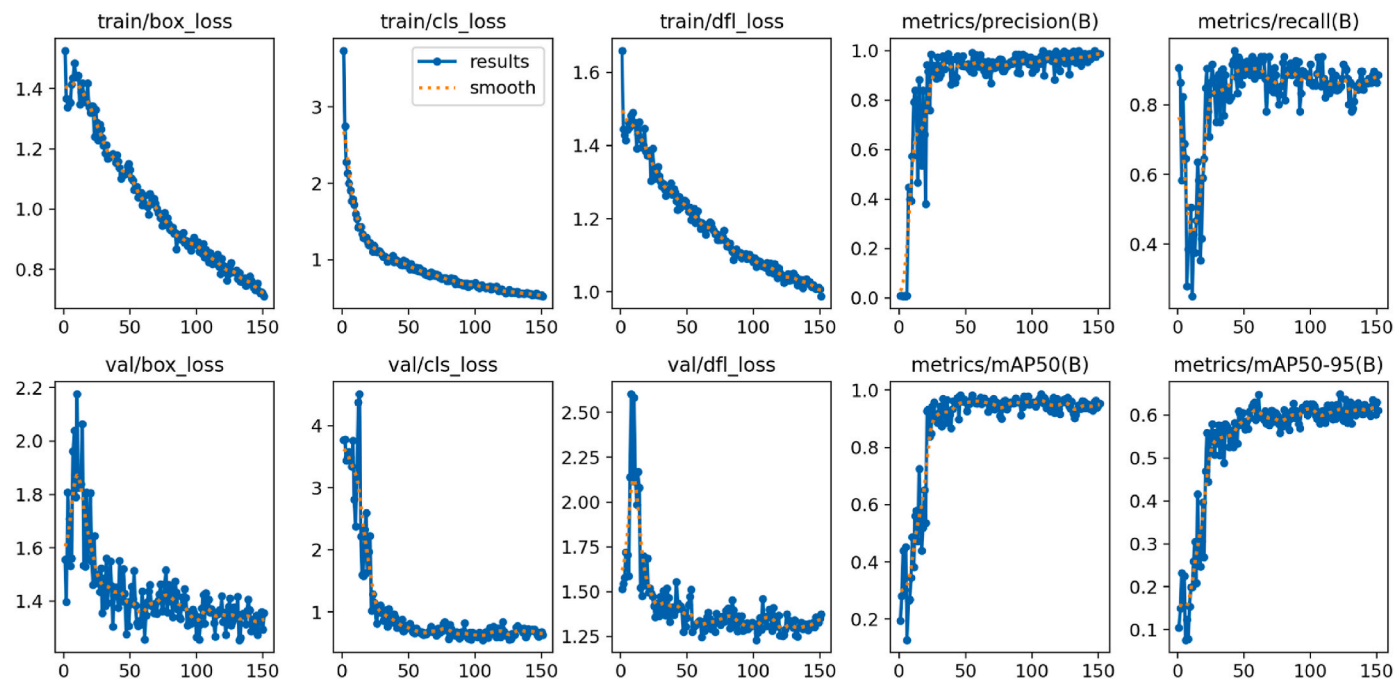


Fig. 6. Training results of YOLOv11n model.

3.3. Performance of YOLOv11n based on testing dataset

The performance evaluation of the YOLOv11n model for bread contamination detection, as shown in Fig. 7, demonstrates promising results across multiple metrics. From the precision-confidence curve, this model exhibited excellent precision at different confidence thresholds, reaching 1.00 precision at a confidence threshold of 0.851. This indicates good reliability in detecting contamination, with a minimal number of false positives. The precision-recall curve confirmed this, which showed a mean average precision (mAP@0.5) of 0.937, indicating that the model efficiently maintained a balance between precision and recall. The F1-confidence curve showed the highest F1 score 0.91 at a confidence threshold of 0.566, suggesting that a good trade-off is sustained between precision and recall, leading to the correct tagging of contamination. A confusion matrix provides deep-diving information, showing 89 contamination classes correctly identified, with 14 misclassifications of the contamination class and seven background misclassifications, representing quite low errors. Based on these results, YOLOv11n can identify bread samples, either contaminated or clean, to provide an instant contamination detection solution that can be used in industrial food-processing applications. Accuracy, precision, recall, and F1-score point toward the efficacy of the model in detecting contamination, while few false-positive errors signal possible improvement through data augmentation or hyperparameter tuning. The results show that lightweight deep-learning models, such as YOLOv11n, can achieve high performance with less computational consumption in the detection of contamination, reinforcing their feasibility for real-world use.

3.4. Prediction confidence of YOLOv11n model

The image presented in Fig. 8 demonstrates the predictive power of the YOLOv11n model when applied to bread contamination detection

through thermal imaging. Fig. 8 (a) shows the actual contamination area labels, whereas Fig. 8 (b) shows the model predictions. The bounding boxes for the areas of contamination marked in both images differ in that the actual labels provide ground truth annotations, whereas the predicted labels show confidence scores given by the model. From the comparison, the YOLOv11n model has good detection of contamination regions, providing a strong confidence measure, as required by the numerical confidence scores shown in the bounding boxes. The model was able to identify the most contaminated spots, thus showing reasonably good localization accuracy. Minor inconsistencies still exist regarding certain misalignments and bounding box placements, as shown in Fig. 8 (a) to 8(b). The model gives lower confidence scores for some contaminations, indicating a slight degree of uncertainty in distinguishing these regions. The overlap between actual and predicted contamination areas by the YOLOv11n model is significant, thus further verifying its usefulness for real-time bread contamination detection, which is necessary for food safety in an industrial setting. Some improvements with the model could be achieved with increased dataset augmentation and hyperparameter tuning, which would further improve precision, including false positives in the detection performance.

3.5. Performance comparison of different YOLO models

From the analysis of YOLO models for detecting contamination (water, oil, grease, and cleaning chemicals) in bread, we can see that the increased complexity of the model does not guarantee improved performance. The lightweight YOLOv11n (2.59 M parameters, 6.4 GFLOPs) attains a mAP50-95 of 0.607 on par with larger models like YOLOv8s (28.6 GFLOPs, mAP50-95: 0.601), and YOLOv9s (27.4 GFLOPs, mAP50-95: 0.601). This level of performance may be a case in which the model designs are grossly overinvested in that their distinction lies not in design complexity but rather in how easy it is to tell one object from

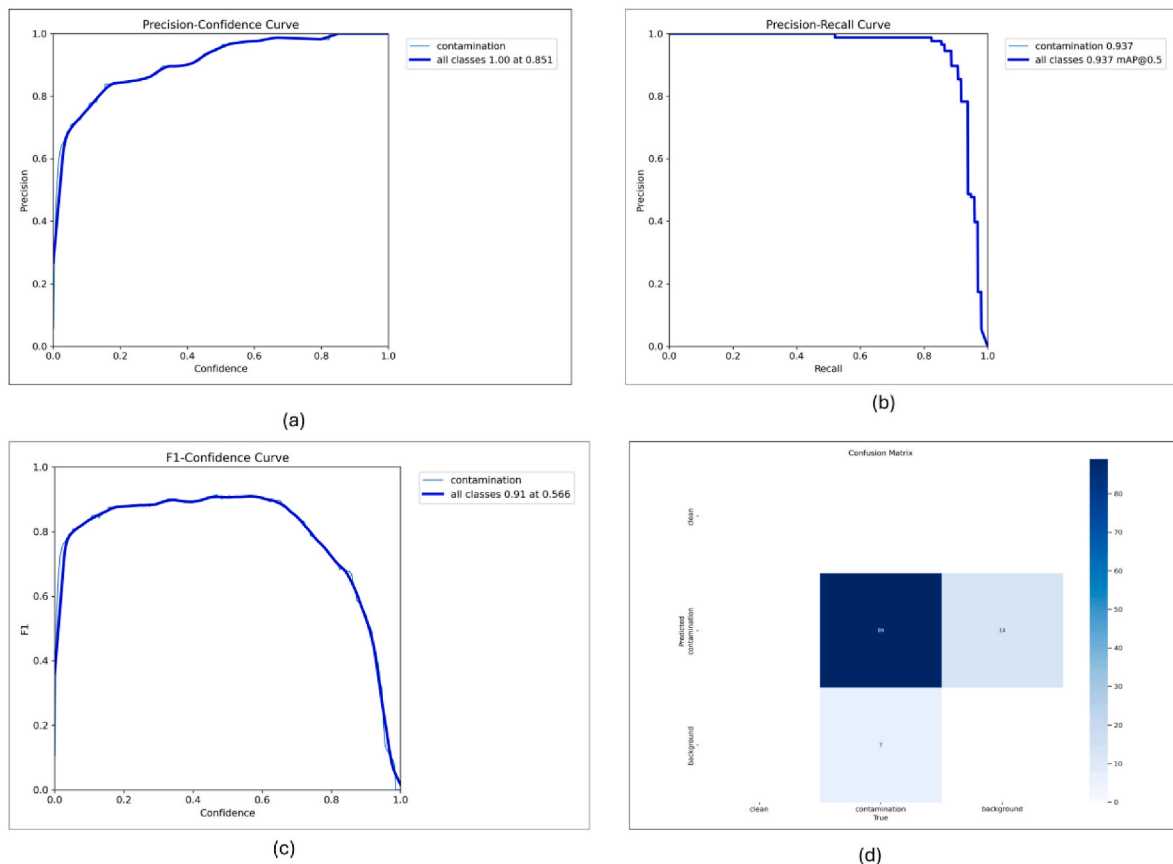


Fig. 7. YOLOv11n model performance on testing.

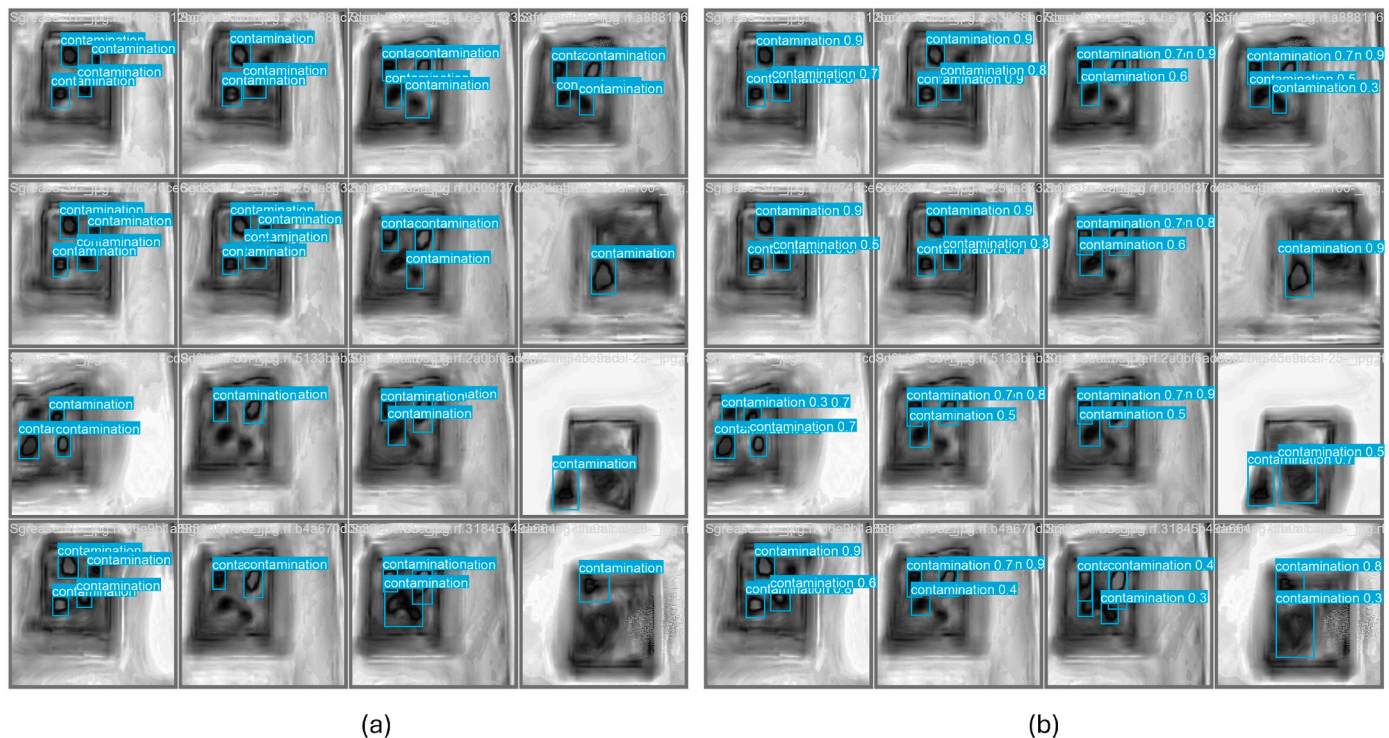


Fig. 8. Prediction confidence of YOLOv11n model.

another-contaminant from bread. With a fast inference speed (3.0 ms/image) and low memory requirements, YOLOv11n provides high efficiency in real-time industrial applications. The other models were trained in a short time ranging from 0.107 to 0.146 h (for 10 epochs) on an NVIDIA Tesla T4 GPU (15 GB memory) with PyTorch 2.5.1, CUDA 12.4, Ultralytics 8.3.70, and Automatic Mixed Precision (AMP); thus, it can be concluded that defect detection is an easy learning task. Complex models seemed to overfit with no performance gains, despite being a deeper architecture. This shows that lightweight models, such as YOLOv11n, afford the best tradeoff between accuracy, computational efficiency, and ability to be field-deployed, especially when working in resource-constrained environments. Table 1 shows the Performance comparison of different YOLO models.

The analysis of training data through different YOLO models, shown in Fig. 9, allows for the identification of some key trends in efficiency, loss convergence, and performance, which can support the conclusion that lighter models, such as YOLOv11n, strike the most acceptable balance between efficiency and performance for contamination detection in bread. In terms of training time, YOLOv11n took the shortest time to complete 10 epochs in approximately 359 s, trailing closely, first by YOLOv5s at 381 s and then by YOLOv8s at 387 s. YOLOv9s, however, was significantly delayed, completing in approximately 501 s, thus marking further contempt on the computational burden of a more complex architecture while garnering any significant improvements in performance. Looking at the loss convergence among the models, YOLOv8s had the most optimal convergence in both box loss (0.90562) and classification loss (0.61157), whereas the classification loss for YOLOv11n was the highest at 0.96013, which still lies within an

acceptable range. In the end, all models produced relatively consistent estimates of direct fine line (DFL) loss of approximately 1.10–1.23, implying that fine performance for detecting object boundaries was present for all of them. Considering their precision and recall performance parameters, the models were considered reliable, with precision values higher than 0.92 across the board and recalls falling between 0.86 and 0.89, as observed. Moreover, they showed strong object detection abilities, with an mAP50 exceeding 0.93, up to 0.95, and reliable performance in contaminant detection, with a consistent mAP50-95 of approximately 0.60–0.61. The models followed an identical learning rate schedule with an initial setting of approximately 0.07 for pg0 and 0.003 for pg1/pg2, with subsequent decay up to 0.001 at the last epoch, which provided stable optimization for all architectures. The evaluation metric plots of the other models are presented in supplementary materials Figs. S1–S5.

3.6. Online model implementation

Fig. 10 Screenshot showing the testing phase of the YOLOv11n model for detecting bread contamination. The thermal imager viewed the spatial temperature distribution across the surface of the bread sample, whereas the heat patterns revealed significant variations, indicating probable sites for contamination. The model output is defined as a bounding box that marks the contaminated area within the thermal view. A confidence score of 97.4 % demonstrated the reliability of the model prediction. The performance metrics extracted from the Roboflow platform expressed mean Average Precision (mAP) with a score of 97.4 %, which showed that the detection was the most successful across

Table 1
Performance comparison of different YOLO models.

YOLO Version	Parameters	Layers	GFLOPs	Training Time (10 epochs)	Final mAP50-95	Inference Speed (ms/image)	Memory (GPU_mem)
YOLOv8s	11,136,374	225	28.6	0.108 h	0.601	4.5 ms	3.88G
YOLOv11n	2,590,230	319	6.4	0.107 h	0.607	3.0 ms	2.37G
YOLOv5s	9,122,966	262	24.0	0.112 h	0.609	4.4 ms	4.28G
YOLOv9s	7,288,182	917	27.4	0.146 h	0.601	7.4 ms	5.10G

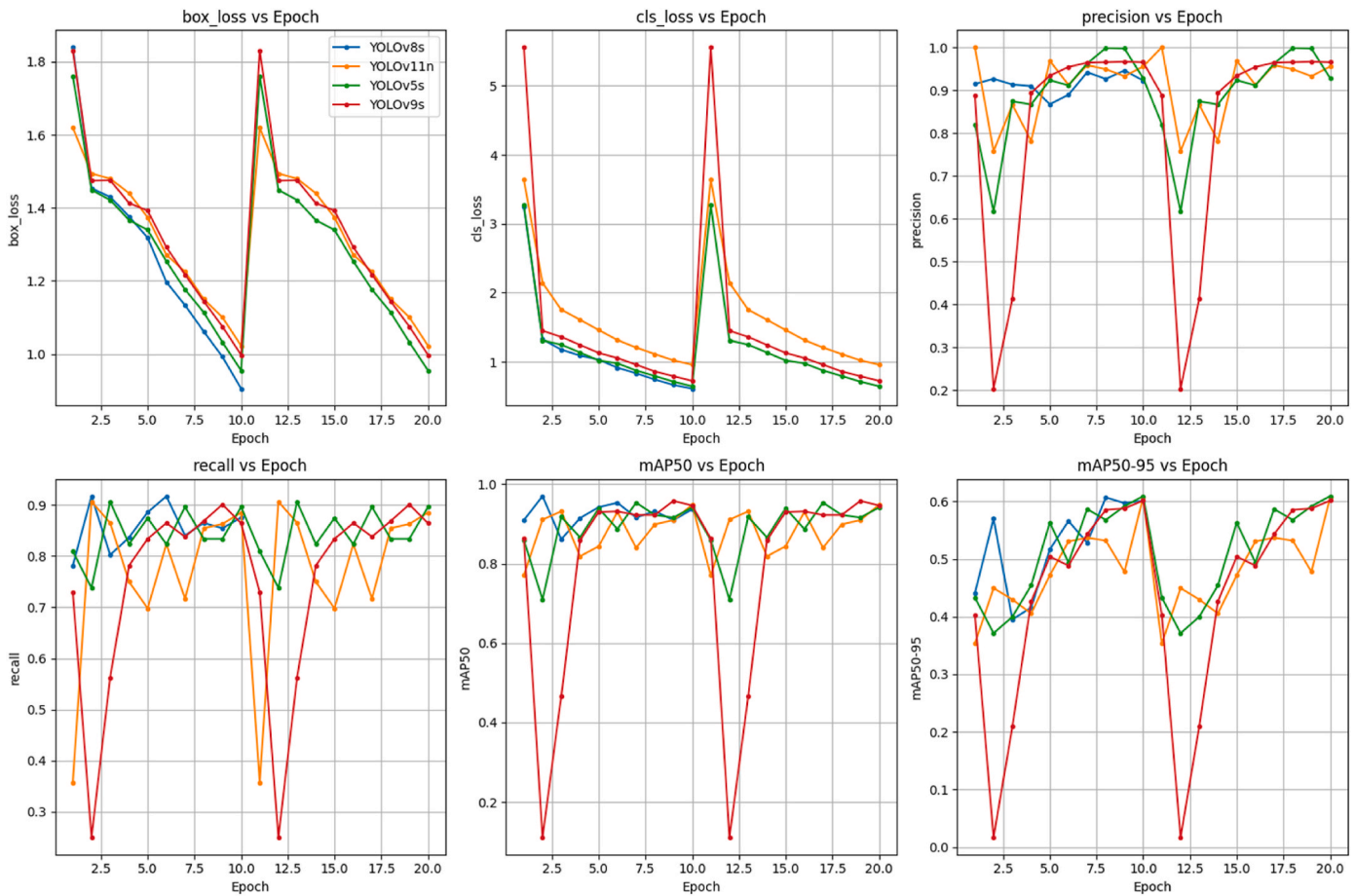


Fig. 9. Comparison of training performance parameters of different YOLO models.

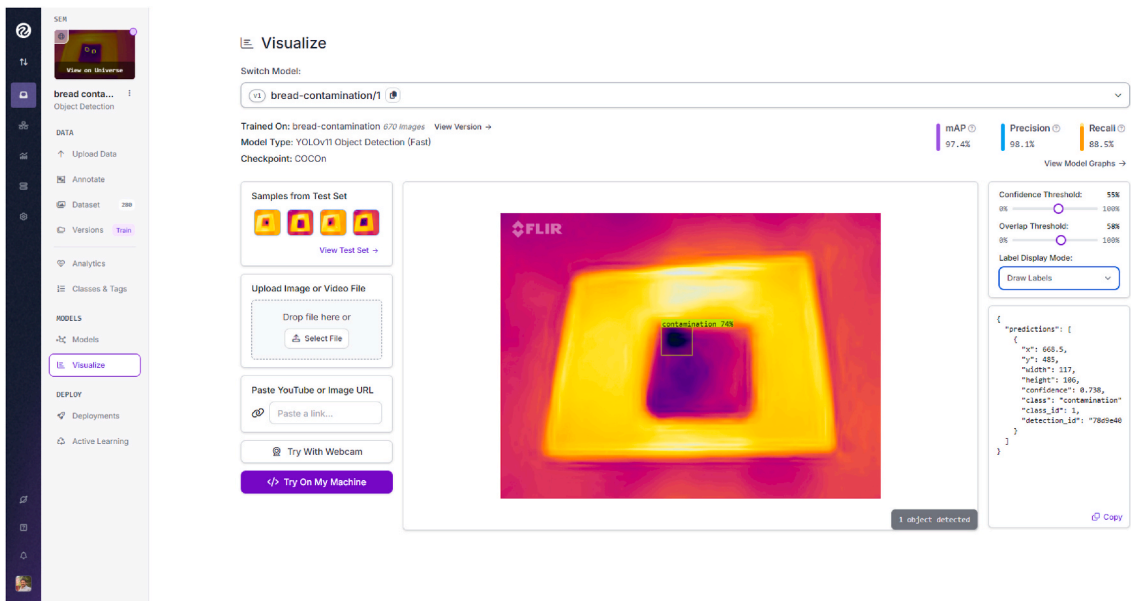


Fig. 10. Screen shot of testing phase of the YOLOv11n object detection model available in roboflow platform.

various contamination conditions. Thus, the established visualization tool was key to the rapid performance assessment and iterative improvement of the detection algorithm throughout the development.

4. Discussion

The effectiveness of multiple YOLO models (YOLOv5s, YOLOv8s, YOLOv9s, and YOLOv11n) in detecting contamination of bread via thermal imaging was studied. The main findings indicate that the

lightweight YOLOv11n model yields the highest trade-off between accuracy and computational efficiency, recording an mAP50-95 score of 0.607, but a much lower computational load than larger models. The results show that different contaminants have different thermal signatures; oil-contaminated areas were found to be the warmest, with water, grease, and chemical contaminants registered at cooler readings. This demonstrates the ability of thermal imaging in food safety applications for precise contamination detection, even when visual inspection fails. The YOLO models proved successful in identifying areas of contamination and could provide a basis for real-time implementation in industrial applications.

Although methods such as microbial culture (Abedi et al., 2020), spectroscopy (Kan et al., 2025), and hyperspectral imaging (Feng et al., 2025) ensure accuracy, they require considerable pre-processing, special equipment, and long processing times, rendering them impractical for real-time applications. Technological developments in recent years have resulted in an increase in sensitive methods, such as cooling quantum dots (CQDs) (Kumar and Rhim, 2024) and electronic nose (E-nose) systems (Sanaeifar et al., 2017). However, these methods are obstructed by poor selectivity, the need for a foul calibration process, and external environmental variability.

YOLO models allow for faster detection by providing real-time classification and localization with high accuracy, thus rendering them much more adaptable and quicker than traditional methods. Unlike an E-nose system that requires lengthy calibration, this system allows for the timely detection of contamination patterns with no reliance on environmental sensitivity. Thus, by combining high accuracy, rapid assessment, and ease of deployment, YOLO-based thermal imaging has emerged in response to the pressing need for state-of-the-art continuous food safety monitoring. Owing to its industrial scalability, YOLO-based thermal imaging is a viable option for contamination detection. Increasing the dataset, particularly with diverse thermal images under varying conditions, can enhance the generalization and accuracy of the model. An accuracy approaching 0.99 may be achieved through advanced data augmentation, fine-tuning with thermal-specific pre-trained weights, optimized hyperparameters, and ensemble techniques. These approaches help the model to better capture the subtle thermal variations associated with different contaminants.

Even though the initial outcomes appeared favorable, certain constraints must be considered. Despite its diversity, the dataset used in this study was limited in terms of size. An increase in the number of contamination scenarios included in the dataset as well as different varieties of bread would aid in producing generalizable models. There were also some misclassifications pertaining to grease and chemical contamination, suggesting that model fine-tuning and data augmentation could enhance discrimination. In addition to improving the prediction and quality of the dataset, the use of an advanced thermal camera is recommended. Future studies should also investigate the possibility of combining multimodal data, which would confer even greater reliability upon detection. Moreover, the computational efficiency of YOLOv11n ensures its suitability for resource-constrained environments, such as small- and medium-scale food processing facilities. The integration of lightweight YOLO models with thermal imaging offers a scalable and practical solution for improving the early stage detection and rejection of contaminated products, thereby reducing food waste and ensuring consumer safety. Additionally, future research may explore deployment in live industrial workflows, integration with spectroscopic or visual sensors, and adaptation to other food categories to broaden their applicability and regulatory compliance.

5. Conclusion

This study demonstrates the implementation of YOLO-based models for object detection in bread contamination detection through thermal imaging. Choosing lightweight models, such as YOLOv11n, would yield the best compromise between accuracy and computing power. The

model attained a confidence level of 0.607 for mAP50-95, while processing at an inference speed of 3.0 ms per image, thereby enabling the solution to be operational for real-time industrial use. The study further demonstrated that the contaminants exhibit different thermal signatures, with oil-affected areas being warmer, while water, grease, and chemical contaminants appear cooler. These findings indicate the potential of thermal imaging in tandem with deep learning for the automatic evaluation of food quality. Although this study proposes a scalable and efficient approach for contamination detection, further studies must be conducted to enhance the model generalization. Expanding the dataset, implementing multimodal imaging techniques, and fine-tuning the model architectures could improve detection accuracy and robustness.

CRedit authorship contribution statement

G. Madasamy Raja: Writing – original draft, Validation, Supervision, Funding acquisition. **P. Pathmanaban:** Writing – review & editing, Writing – original draft, Validation, Conceptualization. **P. Selvaraju:** Writing – review & editing, Methodology, Investigation, Conceptualization. **S. Vanaja:** Software, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jfoodeng.2025.112639>.

Data availability

I have shared the link to my dataset at the attach file [stepThermal-Images-of-Bread-Contamination-Labeled \(Original data\)](#) (Mendeley Data).

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